

Qatar-1b

R-2820

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130703 · created 2026-09-05T02:23:52+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456476.927358 +/- 7.2e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.054
Transit depth (Rp/Rs)^2	2.45 +/- 1.68 [%]
Orbital Inclination (inc)	89.08 +/- 2.92
Airmass coefficient 1 (a1)	0.0281 +/- 0.0049
Airmass coefficient 2 (a2)	2.92 +/- 1.29
Scatter in the residuals of the lightcurve fit is	17.01 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	3.75
Transit Duration (day)	0.0453 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.054 vs 0.1463 (deviation 0.2σ)
Duration fit vs published	0.0453 d vs 0.0692 d (deviation 5.94σ)
Mid-time O–C	0.1 min
Depth SNR	2.9
Residual scatter	17.01 %

Light curve

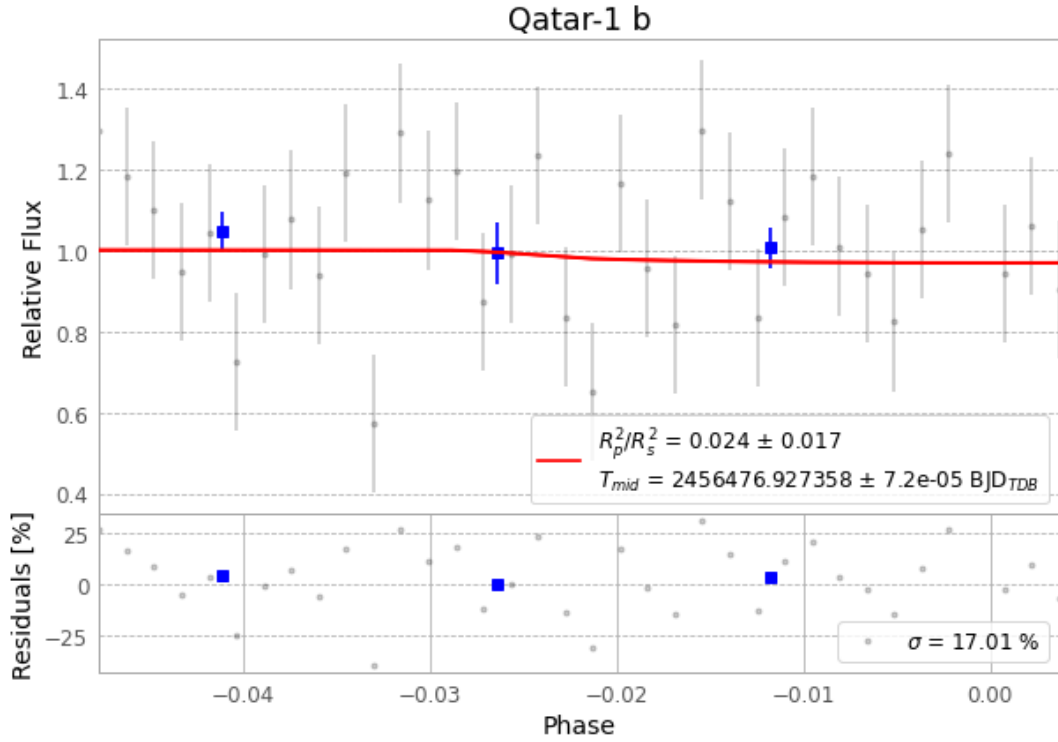


Figure 1 — FinalLightCurve_Qatar-1 b_2013-07-03.png (SHA-256 d4d5968b4ae895f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [530, 351], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eb51ef8b997ba05e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

36 FITS frames (24 MB), Qatar-1130703083612.FITS → Qatar-1130703102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130703.log (217be49577e2...), FinalLightCurve_Qatar-1 b_2013-07-03.png (d4d5968b4ae8...), FinalLightCurve_Qatar-1 b_2013-07-03.csv (cbacbcc464c5...)

Compute resources

Wall time 349.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 02:23Z.